Queen Mary
University of London

EBU5376
Digital Signal Processing

Week 4 - Tutorial

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Tutorial Outline

- Finite Impulse Response (FIR) filter
  - Window functions
- Infinite Impulse Response (IIR) filter
  - Bilinear transform
  - Frequency warping
- Cut-off frequency

Tutorial Question 1

- Finite Impulse Response (FIR) filters can be constructed using the window method. There are three commonly used windows: Rectangular window, Hamming window, Hanning window. The equations of the windows are given as follows:

$$w_{rec}[n] = 1, \qquad -M \leq n \leq M$$

$$w_{ham}[n] = 0.54 + 0.46\cos\left(\frac{n\pi}{M}\right), \qquad -M \leq n \leq M$$

$$w_{han}[n] = 0.5 + 0.5\cos\left(\frac{n\pi}{M}\right), \qquad -M \leq n \leq M$$

Given a filter with an impulse response of $h[n] = [0.50, 0.55, 0.70, 0.55, 0.50], n = -2, \ldots 2$. Apply the Hamming window and the Hanning window to the filter, respectively.

Tutorial Question 2

Given an analog filter with a transfer function of $H_a(s) = \frac{1}{s+1}$, you are going to convert it to a digital filter using a sampling rate of 2 Hz.

i. The Bilinear Transform is a method to obtain the discrete-time approximation of a continuous-time system. State the equation of Bilinear Transform and describe the meaning of each parameter in the equation.

ii. Using Bilinear Transform, obtain the transfer function of the digital filter.

Tutorial Question 3

This question is about Butterworth filters.

i. An analog Butterworth filter is defined with an amplitude response of

$$|H_a(\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}}$$

where ω denotes the frequency, n denotes the order, and ω_c denotes the cut-off frequency. Explain what is meant by the 'order' of a Butterworth filter and how it relates to the filter's transfer function.

ii. Write down the analog transfer function of an order-1 Butterworth filter with a cut-off frequency of $\omega_c = 1$ rad/s.

iii. Using Bilinear Transform, construct a digital filter using the Butterworth prototype defined in part ii. Assume the sampling rate is 4Hz. You don't need to pre-warp the frequency.

Tutorial Question 4

This question asks you to create a digital filter using a Butterworth prototype.

i. Describe the characteristic of the passband response of a Butterworth filter

ii. By using Bilinear Transform, the relationship between the analog frequency Ω and the digital frequency ω is $\Omega = \frac{2}{T}\tan\frac{\omega T}{2}$, where T is the sampling interval. Based on this relationship, describe the disadvantage of Bilinear Transform.

iii. We want to construct a digital filter with a cut-off frequency of 2 rad/s. Derive the cut-off frequency of the analog prototype, assuming the sampling rate is 2Hz.

iv. Using Bilinear Transform, construct a digital filter using the following Butterworth prototype with cut-off frequency denoted by ω_c:

$$H(s) = \frac{1}{\frac{s}{\omega_c} + 1}$$

Assume the sampling rate is 2Hz. The resulting digital filter must have a cut-off frequency of 2 rad/s. Derive the transfer function of the digital filter.

Tutorial Question 5

Figure 1 shows the amplitude response (on linear scale) of a digital filter. Based on this plot, answer the following questions:

i. Describe the filter represented by this amplitude response.

ii. Estimate the cut-off (normalized) frequency from this plot.

iii. Suppose the sampling rate is 4Hz, what is the estimated cut-off frequency in Hz?

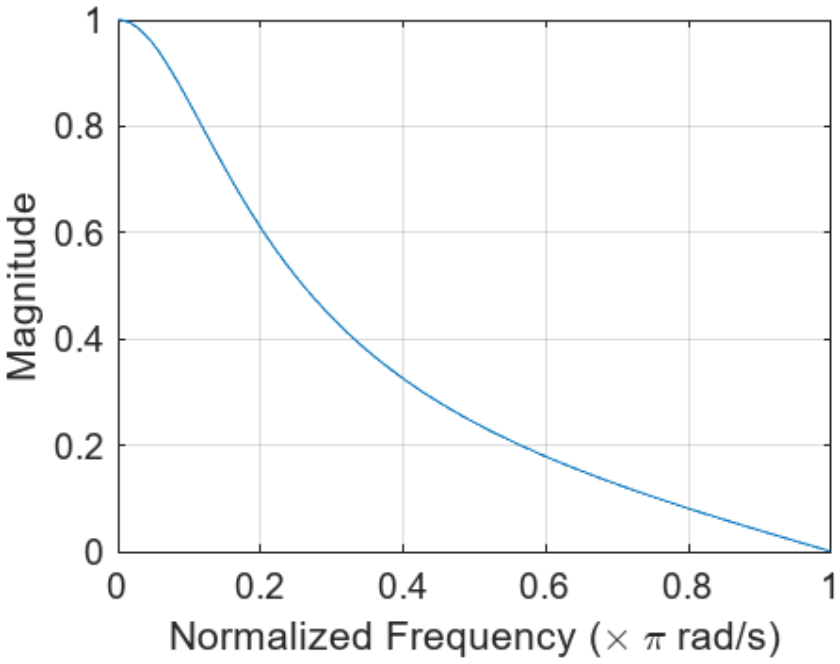

Figure 1

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